Certified support and attachment-preserving product reductions for unknot recognition

Research continuation for the ProveIt project

12 October 2026

Abstract

We develop two reductions at the present boundary between implemented normal-surface operations and a controlled hierarchy algorithm. The algebraic reduction determines the full feasible support of a nonnegative matching cone before ray enumeration. Its certificate consists of a feasible vector and a nonnegative row-space vector with complementary supports. The certificate is checked by exact arithmetic in the original source equations. A direct formulation in these sparse equations has an $O(t^2)$-bit certificate for t tetrahedra, without triangle elimination. We prove an exact quotient theorem for extensions, dual activation constraints, a sparse family invisible to bounded-size combinations of its specified rows, and an intrinsic-dimension bound for conflict branching. The geometric reduction extracts the signed cyclic order of core ports on compressed parallelity bases. Puncturing the boundary at the ports converts ordering into endpoint matching; a second query fixes relative fibre coordinates. Zero gaps are retained. Disc-base components consequently admit explicit linear-size product templates preserving their labelled attachments and the canonical first-cut seams. We provide implementations, independent checkers, source audits, reproducible measurements, and precise integration interfaces. The reductions remove specific sources of wasted work and lost attaching information. The complete sector-cover and patterned-hierarchy bounds needed for a general quasi-polynomial recognizer remain open.

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1 The research question and the source checkpoint

The objective is a recognizer with a proved quasi-polynomial worst-case bound, starting from the existing implementation in `Topology/UnknotRecognition`. This requires controlling both the search for useful surfaces and the topological representation after cutting. A fast first cut is insufficient if the next step must expand the surface or reconstruct its attachments from a large unstructured object. Similarly, fast enumeration in a low-dimensional cone is insufficient if discovering that cone requires an expensive enumeration in a much larger space.

This article addresses these two interfaces. Its mathematical results are stated independently of the implementations. Its experimental claims refer only to completed measurements recorded in the package. In particular, a supplied-sector computation, a parallelity query, and complete unknot recognition are different tasks.

1.1 Precisely what was inspected

The working checkout is

`90e16db798c8d976bbe0c062e07cf20565e0b63a`.

Its relevant `fast/` and `synthesis/` trees agree exactly with the last topology update

`e7b51c4708baefbc5836ef233301a7de3552f86b`,

dated 12 October 2026. The tree identities and individual source hashes are retained in the package manifest. This distinction matters because the repository contains simultaneous work in other mathematical areas.

The source already contains the following ingredients [1]:

- exact supplied-sector matching kernels and complete standard-ray enumeration, including specialized geometry in dimensions at most three;
- inexpensive sufficient forced-zero rules, and a complete feasible-span reduction obtained *after* a complete quadrilateral-ray enumeration;
- compressed normal components, raw cuts in all cell dimensions, and independent source reconstruction;
- full source parallelity bases, fibre twisting, and an inherited complementary core with at most $30t$ handles;
- a general exact recognition fallback, without a proved general quasi-polynomial bound.

The first contribution avoids the expensive prerequisite in the second item. The second makes a particular part of the fourth item's implicit attaching information explicit and usable by a replacement operation. The original source packet remains available as provenance.

1.2 Literature and the meaning of the target

Two primary versions of Lackenby's 2021 announcement state different exponents. The February slides give $n^{O(\log n)}$; the March Oxford handout gives $2^{O((\log n)^3)} = n^{O((\log n)^2)}$ [2, 3]. Both are quasi-polynomial. The repository's attribution of the first bound to the February slides is therefore supported. One should specify the version instead of silently substituting the March exponent.

The July 2026 preprint supplies a hierarchy algorithm and important compressed-decomposition results. Proposition 9.1 bounds the number of iterations by $L(g+1)^L$, in terms of hierarchy length and pattern complexity; its later sections study bounded handle types and certificates [4].

These results motivate the present interfaces. They do not supply the missing global bound for the code studied here.

The linear-algebra foundation below is the theorem of alternatives and rational linear programming [6, 7]. The orbit foundation is the Agol–Hass–Thurston interval-pairing method [5]. We do not claim either foundation, ordinary strict complementarity, or the PL classification of a disc bundle as new. The contributions are the specific complete certificates, transfer statements, obstruction families, compressed port reconstruction, relative replacement specification, and their implementation at this repository checkpoint. Priority of these derived results in the broader literature is not asserted.

1.3 A contribution ledger

Result	What it establishes	Status in this package
Exact support certificate	Every removed coordinate is impossible; every retained coordinate is positive in one feasible vector.	Proved and implemented.
Direct sparse-source LP	An original-source certificate of $O(t^2)$ bits without triangle elimination.	Proved alternative formulation.
Quotient extension and dual activation	When support can be reused, how dimension changes, and why a small column change can still revive a large cone.	Proved; future search interface.
Bounded-row obstruction	Sparse matching-like systems of arbitrarily large raw nullity evade bounded-row sign consequences.	Proved; abstract fixtures.
Dimension-controlled conflict search	A bound exponential in feasible span dimension for homogeneous linear feasibility with quadrilateral exclusions.	Proved; no essential-disc inference.
Cyclic port and fibre recovery	The order, endpoint orientation, gap data, and relative fibre information at core contacts.	Proved and implemented.
Disc product templates	A small PL replacement relative to labelled core attachments and canonical first-cut seams.	Proved; explicit templates emitted.
General quasi-polynomial recognition	Complete useful-surface discovery and a controlled patterned hierarchy.	Open.

2 Normal coordinates and the precise algebraic interface

Let t denote the number of tetrahedra. A standard normal vector has four triangle and three quadrilateral coordinates in each tetrahedron. Across each paired face, the three types of normal arcs must match. Write these homogeneous integer equations as

$$M_{\triangle} a + M_{\square} q = 0.$$

Nonnegativity is required. In addition, at most one quadrilateral type can be nonzero in each tetrahedron. Throughout the supplied-sector interface, the allowed types Σ already satisfy this compatibility restriction. A coordinate may be allowed and still be forced to zero.

The existing source kernel eliminates triangle differences, retaining vertex-link freedom, and obtains a homogeneous rational system

$$A_\Sigma q = 0, \qquad q \geq 0.$$

The relevant cone is

$$C_\Sigma = \ker(A_\Sigma) \cap \mathbb{R}^k_{\geq 0}, \qquad k = |\Sigma|.$$

Its *raw nullity* is $d = k - \operatorname{rank} A_\Sigma$. Its *effective dimension* is

$$s = \dim \operatorname{span} C_\Sigma.$$

The two can be very different: a linear kernel contains signed directions that may be absent from the nonnegative cone.

For this finite compact source model, a quadrilateral solution satisfying the elimination equations admits triangle potentials. Adding sufficiently large vertex links makes all triangle coordinates nonnegative. This source-kernel property is already implemented and is independently checkable through the original face equations. It is not an assertion about arbitrary ideal or spun-normal coordinate systems.

2.1 Why a complete support certificate changes the cost

Let

$$S = \{j : \text{there is } q \in C_\Sigma \text{ with } q_j > 0\}.$$

After proving that all coordinates outside S vanish, the restricted cone has a strictly positive point. It therefore spans its whole restricted kernel:

$$\operatorname{span} C_\Sigma \cong \ker A_{\Sigma,S}.$$

Consequently the existing low-dimensional ray machinery can be dispatched using s, without first enumerating in raw dimension d.

This is a substantive difference from merely assigning numerical zeros to small coordinates in an LP solution. An LP vertex typically has many zero coordinates that are positive elsewhere in the cone. The certificate must prove which zeros are *unavoidable*, and must prove that its retained set is complete.

2.2 A certificate directly in source equations

The implementation can bind the support statement to the source without asking the verifier to trust the producer's elimination:

1. supply an integral standard normal vector $v = (a, q)$ whose quadrilateral support is exactly S;
2. supply coefficients λ on the original face-matching rows;
3. verify
$$M_\triangle^\mathsf{T} \lambda = 0, \qquad y_\Sigma = (M_\square)_\Sigma^\mathsf{T} \lambda \geq 0;$$
4. verify $q \geq 0$, the original matching equations, and $q_j + y_j \geq 1$ for every allowed quadrilateral coordinate.

Here the nonnegativity requirement concerns only allowed columns; coefficients on disallowed quadrilateral columns are unrestricted. For any source-feasible standard vector (a', q') in the stated sector, q' vanishes outside Σ, and

$$0 = \lambda^\mathsf{T}(M_\triangle a' + M_\square q') = y_\Sigma^\mathsf{T} q'_\Sigma.$$

Thus every coordinate on which y is positive vanishes in every source-feasible vector. The supplied q witnesses simultaneous positivity on the complementary set. Triangle cancellation in the dual is essential: without it, triangle coordinates could compensate for a positive quadrilateral term.

Only columns in the stated sector are considered. A certificate for one sector does not justify deleting a type in a larger sector unless the extension conditions proved below are checked.

3 Complete support by one primal–dual feasibility problem

Let $A \in \mathbb{Q}^{m\times k}$ and write

$$C(A) = \{x \in \mathbb{R}^k : Ax = 0,\ x \geq 0\}, \qquad S(A) = \{j : \text{some } x \in C(A) \text{ has } x_j > 0\}.$$

The complement $F(A) = \{1, \dots, k\} \setminus S(A)$ is the set of coordinates forced to zero. All inequalities between vectors are coordinatewise. A dual exposing vector means a vector $y = A^T\lambda \geq 0$ whose zero set is exactly $S(A)$.

Theorem 3.1 (Complete support from a joint linear certificate)**.** *The linear system*

$$Ax = 0, \qquad y = A^T\lambda, \qquad x \geq 0, \quad y \geq 0, \qquad x + y \geq \mathbf{1} \tag{1}$$

is feasible for every rational matrix A. Every feasible solution has

$$\operatorname{supp}(x) = S(A), \qquad \operatorname{supp}(y) = F(A).$$

In particular, a verifier needs only exact matrix arithmetic and sign tests, rather than access to the optimization routine producing the certificate.

Proof. First prove the coordinate alternative. Let a_j be column j of A. There is an element of $C(A)$ with positive jth coordinate if and only if

$$A_{-j}u = -a_j, \qquad u \geq 0$$

is feasible: scale the positive coordinate to one. If this system is infeasible, Farkas' lemma supplies $\lambda^{(j)}$ such that

$$A_{-j}^T\lambda^{(j)} \geq 0, \qquad a_j^T\lambda^{(j)} > 0.$$

Thus $A^T\lambda^{(j)} \geq 0$, with a positive jth coordinate. Conversely, such a vector forces $x_j = 0$ for every $x \in C(A)$, since

$$0 = (Ax)^T\lambda^{(j)} = x^TA^T\lambda^{(j)}$$

is a sum of nonnegative terms. This also checks the sign in the Farkas alternative.

For each $j \in S(A)$ choose a primal witness and scale it so its jth coordinate is at least one. Their sum x is positive on every coordinate of $S(A)$ and zero outside $S(A)$. For each $j \in F(A)$ choose the dual witness just constructed and scale it so its jth coordinate is at least one. Sum the corresponding multipliers to obtain λ, and set $y = A^T\lambda$. Every summand of y vanishes on $S(A)$ by orthogonality with feasible primal witnesses. Consequently $x + y \geq \mathbf{1}$, proving existence. Empty sums cause no difficulty.

Now consider any solution of (1). We have $x^Ty = 0$, so nonnegativity gives $x_jy_j = 0$ for each j. The last inequality ensures that exactly one of x_j, y_j is positive. Every positive x_j witnesses $j \in S(A)$, while every positive y_j forces $j \notin S(A)$. This proves the claimed partition. □

The theorem is a specialization of classical alternatives and strict complementarity for the nonnegative orthant. Its use here is a complete replacement for sufficient one-row or two-row sign propagation.

3.1 Polynomial bit complexity, including redundant rows and lineality

Proposition 3.2 (Small exact certificates)**.** *After clearing row denominators, suppose A is integral with $|A_{ij}| \le H$, where $H \ge 1$. System (1) has a rational solution of polynomial encoding length in $m, k, \log H$. More explicitly, put $v = 3k + 2m$ and let $\rho \le m + 2k$ be the rank of the equality matrix in the standard-form system below. One may choose a common denominator and all primal and dual variable numerators of magnitude at most*

$$D = \rho! H^{\rho},$$

except that a multiplier λ_i obtained as a difference of two nonnegative variables may require the bound $2D$. The convention $D = 1$ applies if $\rho = 0$. In particular the total number of bits is $O(v\rho(\log(H+1) + \log(\rho+1)))$, up to ordinary bookkeeping.

Proof. Split $\lambda = \lambda^+ - \lambda^-$ and introduce a nonnegative surplus vector z. All variables in

$$Ax = 0, \qquad A^T\lambda^+ - A^T\lambda^- - y = 0, \qquad x + y - z = \mathbf{1}, \qquad x, y, z, \lambda^+, \lambda^- \ge 0$$

are nonnegative. This is a feasible standard-form rational system $Mu = b$, with integral coefficients bounded by H and with entries of b equal to zero or one. Remove redundant equality rows, leaving ρ independent rows. Row removal means choosing a subset of the original rows; it does not enlarge the coefficient bound. Since the system is feasible, the removed equations follow from those retained with the correct right-hand sides.

For completeness, a nonempty standard-form feasible set has a basic feasible solution even when the original free-multiplier formulation has lineality. Choose a feasible vector with minimum positive support. If the columns of M on that support were dependent, perturb within their dependence in a direction and for a distance that makes one positive coordinate zero while keeping all others nonnegative. This preserves the equalities and contradicts minimality. The support columns are therefore independent. Extend them to a basis of ρ columns, assigning zero to any added coordinates. This produces a basic feasible solution. Equivalently, the nonnegative standard-form feasible set contains no affine line, despite potentially unbounded recession directions.

Cramer's rule on the corresponding $\rho \times \rho$ nonsingular integer matrix bounds its determinant and all numerator determinants by $\rho! H^{\rho}$. The determinant is a nonzero integer. This proves the bounds and the claimed encoding length. Recombining the split multipliers introduces at most the stated factor two.

For rational A, clearing denominators row by row has polynomial bit cost in its input encoding. If $A' = RA$ is the resulting integer matrix, where R is diagonal, a multiplier μ for A' corresponds to the source multiplier $R\mu$ for A. This transformation again has polynomial encoding length. □

Classical rational linear-programming algorithms therefore find such a certificate in polynomial bit time. This statement concerns the mathematical LP primitive; it does not assert a polynomial pivot bound for a chosen simplex implementation. Clearing a common positive denominator in a joint certificate also produces integral x, y, λ: the inequality $x + y \ge \mathbf{1}$ becomes stronger after this scaling.

4 The support restriction is the intrinsic linear span

Proposition 4.1 (Effective dimension)**.** *Let $S = S(A)$. Extension by zero gives a cone isomorphism*

$$C(A_S) \longrightarrow C(A),$$

and

$$\operatorname{span} C(A_S) = \ker A_S, \qquad \dim \operatorname{span} C(A) = |S| - \operatorname{rank} A_S.$$

Proof. All deleted coordinates vanish on $C(A)$, so restriction and extension by zero are inverse linear maps on the cones. There is a vector $p \in \ker A_S$ with $p > 0$. Given $u \in \ker A_S$, choose $\epsilon > 0$ small enough that $p \pm \epsilon u \geq 0$. Both of these vectors belong to $C(A_S)$, and their difference is $2\epsilon u$. This proves the span assertion. Rank-nullity gives the formula. □

Extension by zero also preserves the complete face lattice, extreme rays, linear objectives and the integral lattice in these coordinate spaces. More generally it preserves any further conditions on the retained coordinates. When applying this to normal surfaces, one must still justify the relationship between the matching cone used for support computation and the actual standard-coordinate source cone. Forced zeros in a relaxation safely remain forced in every geometric subcone; maximality in a relaxation does not establish maximality after quadrilateral exclusions or other geometric restrictions.

5 Exact reuse when allowed columns are deleted or added

Throughout this section, all columns belong to one fixed matrix of source constraints. The results do not authorize reuse across a Pachner move or any other change to the source equations without a separate transport proof. Write $C_I = C(A_I)$ for an allowed column set I. Let $S = S(A_I)$, $F = I \setminus S$, and choose a complete exposing multiplier λ such that

$$y_i = a_i^T \lambda = 0 \ (i \in S), \qquad y_i > 0 \ (i \in F).$$

5.1 Deletion and lifting a smaller dual certificate

Proposition 5.1 (Restriction and dual lifting)**.** *For $I' \subseteq I$, every coordinate of $F \cap I'$ remains forced zero. Complete support for I' may therefore be computed using only $R = S \cap I'$. If I' contains S, the original support and effective dimension are unchanged. In general the effective dimension cannot increase.*

Suppose η is a complete exposing multiplier for the smaller cone $C(A_R)$, and put $v_i = a_i^T \eta$ for all $i \in I'$. Define

$$\alpha = 1 + \max_{i \in F \cap I'} \frac{\max(0, -v_i)}{y_i},$$

where the maximum of an empty set is zero. Then $\eta + \alpha\lambda$ is a complete exposing multiplier on I'. All these operations preserve polynomial certificate bit length.

Proof. The restriction of y is still a nonnegative row-space consequence, so its positive coordinates remain zero on every feasible vector. This gives the cone isomorphism with $C(A_R)$. The cone for I' is a coordinate face of the old cone, embedded with zeros, so its dimension cannot increase. If only old forced coordinates are deleted, the cones are identical under that embedding.

On R, the old exposing vector is zero, so the new exposing vector equals v. On $F \cap I'$, its coefficient is $v_i + \alpha y_i \geq y_i > 0$. Its zero set is therefore precisely the new maximal support. Rational maxima, products and quotients of the displayed certificate entries have polynomial bit length. □

5.2 Which added columns can invalidate an exposing vector?

Theorem 5.2 (Dual sign test for extensions)**.** *Add columns indexed by E, disjoint from I, and let*

$$w_e = a_e^T \lambda, \qquad E_+ = \{e : w_e > 0\}, \quad E_0 = \{e : w_e = 0\}, \quad E_- = \{e : w_e < 0\}.$$

Every $q \in C_{I \cup E}$ satisfies the exact balance identity

$$\sum_{i \in F} y_i q_i + \sum_{e \in E_+} w_e q_e = \sum_{e \in E_-} (-w_e) q_e. \tag{2}$$

Consequently, positive use of an old forced coordinate or of a column in E_+ requires positive use of at least one column in E_-.

If $E_- = \varnothing$, every coordinate in $F \cup E_+$ is still forced zero, and the expanded cone is isomorphic to $C(A_{S \cup E_0})$. Writing $d = \dim \operatorname{span} C_I$ and d' for the corresponding dimension after extension, one has

$$d \le d' \le d + |E_0|.$$

If every added column has strictly positive w_e, the old cone and support remain unchanged.

Proof. Apply λ^T to the matching equation for q and move its negative terms to the other side. This proves (2). Every summand on both sides is nonnegative, giving the necessity statement and, when E_- is empty, all asserted forced zeros. Old feasible vectors extend by zero, so $d' \ge d$. Since $\dim \ker A_S = d$, adding $|E_0|$ columns increases the kernel dimension by at most $|E_0|$. The new feasible span lies in that kernel, yielding the upper bound. The last assertion follows when E_0 is empty. □

Example 5.3 (One added column can cause arbitrary revival)**.** Let $A_I = (1, \dots, 1)$ have r columns. Its nonnegative kernel is zero. Add the single column -1. The new cone is

$$\{(u_1, \dots, u_r, t) \ge 0 : t = u_1 + \cdots + u_r\},$$

which has all $r + 1$ coordinates in its maximal support and dimension r. The old multiplier $\lambda = 1$ evaluates negatively on the added column. Thus no dimension bound depending only on the old effective dimension and the number of added columns can hold without additional hypotheses. In particular the nonnegative sign hypothesis above is substantive.

5.3 Contracting the live core

Theorem 5.4 (Quotient extension and exact dimension sequence)**.** *Let S be a column set admitting $p > 0$ with $A_S p = 0$. Set $V = \operatorname{im} A_S$, and choose an exact linear map Q with $\ker Q = V$. For any allowed set $J \supseteq S$, put*

$$R = J \setminus S, \qquad B = Q A_R.$$

The coordinate projection onto R maps $C(A_J)$ surjectively onto $C(B)$. If $U = S(B)$, then

$$S(A_J) = S \cup U.$$

There is a short exact sequence of real vector spaces

$$0 \longrightarrow \ker A_S \xrightarrow{u \mapsto (u,0)} \operatorname{span} C(A_J) \xrightarrow{\pi_R} \operatorname{span} C(B) \longrightarrow 0.$$

Hence the effective dimensions satisfy the exact additive identity

$$\dim \operatorname{span} C(A_J) = \dim \ker A_S + \dim \operatorname{span} C(B).$$

All matrix constructions can be performed in polynomial rational bit time. The theorem includes $S = \varnothing$, with $V = 0$ and a vacuous positive vector.

Proof. First observe that the columns in S positively generate their entire linear span: if $A_S u = v$ for signed coefficients u, then $A_S(u + cp) = v$, and $u + cp \geq 0$ for all sufficiently large c. Thus $\mathrm{cone}(A_S) = V$.

If $(x, z) \in C(A_J)$, applying Q gives $Bz = 0$, so the projection is well-defined. Conversely, let $z \geq 0$ satisfy $Bz = 0$. Then $A_R z \in V$, so choose signed u with $A_S u = -A_R z$. For sufficiently large c, $u + cp \geq 0$ and $(u + cp, z) \in C(A_J)$. This proves surjectivity on the cones. Every coordinate of S remains feasible by embedding the original positive vector; surjectivity determines exactly which coordinates of R are feasible. This proves the support identity.

The injection in the displayed sequence is well-defined because $\ker A_S = \operatorname{span} C(A_S)$ and $C(A_S)$ embeds into $C(A_J)$. Surjectivity of the last arrow follows by taking spans of the already established cone surjectivity. A vector in the middle space with zero R coordinates has form $(u, 0)$, and its matching equation says $A_S u = 0$. Conversely every such vector belongs to the image of the first arrow. This proves exactness and therefore the dimension formula.

An exact basis of the annihilator of $\operatorname{im} A_S$ provides Q. Standard rational elimination and its minor bounds give the stated construction and bit bounds. □

This quotient construction preserves the linear nonnegative cone. If J contains mutually incompatible quadrilateral types, the lift by cp may use incompatible types; the theorem then describes a safe matching relaxation and does not itself solve the compatibility problem. For a compatible fixed sector, this particular obstruction does not occur, although the separate geometric verification remains.

6 Sparse families separating raw and effective dimension

Theorem 6.1 (Cyclic obstruction to bounded-row sign propagation)**.** *Fix integers $m \geq 3$ and $s \geq 1$. Use nonnegative variables $p_1, \ldots, p_m, z_1, \ldots, z_m, t_1, \ldots, t_{s-1}$, with cyclic indices on the z variables, and impose*

$$p_i + z_i - z_{i+1} = 0 \qquad (1 \leq i \leq m).$$

In the displayed variable ordering the matrix is already in reduced row-echelon form. Its rank is m and its raw matching nullity is $m + s - 1$. Its nonnegative cone has maximal support consisting of all z and t coordinates and has effective dimension s.

Every nonzero nonnegative linear combination of the displayed rows uses all m rows. Here "nonnegative linear combination" means a row combination whose resulting coefficients are all nonnegative; the combination's multipliers are not assumed nonnegative in advance. Thus any sign-forcing method examining only spans of at most $h < m$ of these rows makes no first step and therefore no progress.

Proof. The p columns form the identity matrix, proving the rank and RREF claims. Sum all equations to obtain $\sum_i p_i = 0$. Nonnegativity forces $p_i = 0$ for every i, after which the equations require all z_i to be equal. The common value and the $s - 1$ free t values parameterize the cone as $\mathbb{R}^s_+$. Setting the common value and all free values to one proves the support statement.

For a row combination with multipliers λ_i, the p_i coefficient is λ_i, so a nonnegative resulting row requires $\lambda_i \geq 0$. Its z_i coefficient is $\lambda_i - \lambda_{i-1}$. All these coefficients are nonnegative and their sum is zero. They are therefore all zero, forcing all λ_i to be equal. A nonzero nonnegative result consequently has every multiplier strictly positive and uses all m rows. No bounded-row sign step is initially available, so there are no earlier forced zeros from which an iterative closure could begin. □

An exact joint certificate is particularly small:

$$x_p = 0, \quad x_z = \mathbf{1}, \quad x_t = \mathbf{1}, \qquad \lambda = \mathbf{1}, \qquad y_p = \mathbf{1}, \quad y_z = y_t = 0.$$

Every row has three nonzero entries, and every matrix entry is in $\{-1, 0, 1\}$. For fixed s, the raw dimension tends to infinity while the complete support reduction exposes a fixed-dimensional feasible cone.

The obstruction is deliberately relative to the specified row family, including its RREF presentation. It is not invariant under arbitrary change of row basis: a basis containing the sum of all rows would immediately reveal the forced zeros. That dependence is exactly why fixed one-row and two-row closure is insufficient.

Example 6.2 (A bounded-coefficient variant with effective dimension zero)**.** Replace the cyclic equations by

$$p_i - z_i + 2z_{i+1} = 0.$$

Their sum forces all p_i, z_i to vanish, so the nonnegative cone is zero despite raw nullity m. A nonnegative row consequence has $\lambda_i \geq 0$ and $\lambda_i \leq 2\lambda_{i-1}$. If one multiplier vanishes, these cyclic inequalities force every multiplier to vanish. Thus every nonzero nonnegative consequence again uses all m rows. Appending s unconstrained nonnegative columns gives raw nullity $m + s$ and effective dimension s, for any $s \geq 0$.

These are structural examples for abstract rational matching systems. No realization by triangulated knot exteriors, or by the repository's actual source-derived sector equations, is asserted. They establish an algebraic separation and supply regression cases; a geometric realization theorem would be additional research.

7 Conflict branching controlled by intrinsic dimension

Theorem 7.1 (A dimension bound for compatibility search)**.** *Let A and ℓ be rational, and let $\mathcal{P}$ be a set of forbidden coordinate pairs. Consider the decision problem*

$$Ax = 0, \quad x \geq 0, \quad \ell^T x > 0, \qquad x_i x_j = 0 \quad \text{for every } \{i, j\} \in \mathcal{P}.$$

Put $d = \dim \operatorname{span} C(A)$, and let N be the bit length of the rational input and pair list. Following complete support restriction, this problem can be solved using at most $2^{d+1} - 1$ additional linear-feasibility queries, each of polynomial bit complexity in N. Its total bit running time is $2^{O(d)} \operatorname{poly}(N)$. In particular $d = O(\log N)$ gives a polynomial bound, and $d = O((\log N)^2)$ gives $N^{O(\log N)}$.

Proof. Use homogeneity to replace $\ell^T x > 0$ by $\ell^T x \geq 1$. At a search node retain the matching equations, nonnegativity and the coordinate-zero equations selected along its path. Query feasibility with $\ell^T x \geq 1$. An infeasible node is rejected. If a feasible witness has no forbidden pair positive, it solves the problem. Otherwise choose a forbidden pair $\{i, j\}$ whose two coordinates are positive in this witness and branch on $x_i = 0$ and $x_j = 0$. Every compatible feasible vector belongs to at least one child, so the branching is complete.

Let C_v be the homogeneous matching cone at a feasible node. The coordinate functionals x_i and x_j are both nonzero on $\operatorname{span} C_v$, as witnessed by the positive conflicting coordinates. Each child cone is contained in the kernel of one of these nonzero functionals. Consequently each child has span dimension at most $\dim \operatorname{span} C_v - 1$; the child is also a proper coordinate face of C_v. At depth d every remaining cone is zero, so its normalized LP is infeasible. A binary tree of depth at most d has at most $2^{d+1} - 1$ nodes.

All node systems consist of the original input and coordinate-zero conditions, so their input bit lengths remain polynomial in N. Rational linear programming supplies the required polynomial cost per query. The displayed complexity consequences follow directly. □

It is enough to compute complete support once at the root: its restricted kernel has dimension exactly d. Recomputing support at children can improve practical pruning and dimension estimates but is unnecessary for the bound. Similarly, a zero-dimensional homogeneous cone can be rejected before an LP call, improving constants but not the theorem's stated bound.

Quadrilateral exclusions are represented by the three forbidden pairs among the three quadrilateral types of each tetrahedron. The theorem is a refinement of familiar LP-guided compatibility branching, not an assertion that such branching was previously unknown. Its useful parameter is the intrinsic matching-cone dimension exposed by the complete support certificate.

This theorem is an algebraic existence result. An arbitrary positive linear objective does not imply an essential normal disc. Even when the objective is Euler characteristic in genuine standard normal coordinates, vertex links, spheres, disconnected components and boundary-parallel discs require the applicable established geometric arguments and verification steps. Nor has a logarithmic-squared bound on d for arbitrary knot-exterior inputs been proved here.

8 Implementing source support certificates

8.1 Producer and independent checker

The new producer accepts either an explicit rational matrix or a source triangulation with compatible allowed quadrilateral types. In the generic interface the mathematical certificate is a pair (x, λ). In the source interface it is a normal vector together with coefficients on the source face equations. The independent checker verifies exact matrix products, signs, support coverage, and source identity. It does not accept a numerical LP status as a proof.

The current source validator accepts finite, connected, compact, orientable 3-manifold triangulations with exactly one torus boundary component. It checks this source class directly; it does not establish provenance from a knot diagram. The abstract rational-cone theorems have no such topological restriction. The same source-class restriction applies to the geometric APIs used below.

The automatic search path uses floating-point linear programming to propose a support partition and candidate witnesses. Exact reconstruction then solves the corresponding rational equations; the result is accepted only if the full certificate passes. An exact rational simplex path is also available. The implementation details and tested solver modes are recorded with the experiments.

This distinction separates three claims that are easy to confuse:

- a complete mathematical certificate always exists;
- rational linear programming admits polynomial-time algorithms;
- the particular numerical or simplex producer in this package has no claimed polynomial worst-case running time.

The verifier is independent of that last issue. Failure to find or reconstruct a certificate produces Inconclusive, not an exclusion.

8.2 A direct sparse-source formulation

Corollary 8.1 (A direct source LP with small coefficients)**.** *For a finite triangulation with t tetrahedra and a compatible set of $k \leq t$ allowed quadrilateral types, maximal quadrilateral support has an original-source primal–dual certificate of $O(t^2)$ bits, obtainable by rational linear programming without triangle elimination.*

Proof. Retain all $4t$ triangle columns and the k allowed quadrilateral columns in the original matching matrix M_Σ. It has $p \leq 5t$ columns, $m \leq 6t$ rows and entries of absolute value at most 2: each paired face contributes three equations, each comparing a triangle-plus-quad arc count on its two sides. Every row has at most four nonzero entries; every column has at most four, since a triangle has three face arcs and a quadrilateral four. Self-gluings can identify or cancel these incidences and do not increase their number. The vector $u = (\mathbf{1}_{4t}, 0_k)$ satisfies $M_\Sigma u = 0$. Hence every triangle coordinate belongs to maximal support, and every $y = M_\Sigma^\mathsf{T}\lambda \geq 0$ satisfies $0 = u^\mathsf{T} y$, forcing all its triangle coefficients to vanish. The joint support LP therefore directly gives the required source dual and a nonnegative standard primal vector in the stated sector. Its standard form has at most $3p + 2m \leq 27t$ nonnegative variables and equality rank $\rho \leq m + 2p \leq 16t$. Its three row types have at most 4, 9 and 3 nonzero entries, all of absolute value at most 2; their Euclidean norms are therefore at most 6. Choose independent original equality rows and a basic feasible solution as above. The square basis matrix has determinant of absolute value at most 6^ρ by Hadamard's inequality. In a Cramer numerator, replacing a column by the right-hand side in $\{0, 1\}^\rho$ raises each squared row norm by at most one, so every such numerator has absolute value at most 7^ρ. All reduced denominators divide the same basis determinant; clearing the primal denominators therefore preserves the numerator bound. Taking the difference of split dual multipliers introduces at most a factor of two. Thus every certificate entry has $O(t)$ bits, and its $O(t)$ entries require $O(t^2)$ bits in total, including row and column indices. Zero extension restores the primal's disallowed quadrilateral coordinates; the multipliers already refer to the original source rows. □

This is a certificate-existence result and an alternative theoretical LP formulation. The producer benchmarked in this package uses the quadrilateral quotient. The full standard kernel retains vertex-link directions, so its dimension must not replace the effective quadrilateral dimension used for low-dimensional dispatch; that dispatch still uses the quotient. The witness bound is an existence bound, not a bound on arbitrary solver outputs or a performance claim for an implemented direct-source solver.

8.3 The opt-in sector wrapper

After successful verification, the wrapper rebuilds the existing source kernel on the retained allowed types. This preserves the exact matching geometry and canonical lifting behavior. It then calls the ordinary supplied-sector discovery machinery.

For a positive result, the disc certificate still concerns the actual source triangulation and full normal coordinates. For a negative supplied-sector result, the support certificate and the reduced-sector exhaustion must both be retained. The first proves that every feasible vector in the requested sector lies in the reduced sector. The second proves that the reduced sector has been completely searched. Together they establish coverage of the originally requested sector. A capped ray search does not satisfy the second premise.

This package adds an explicit opt-in interface. It does not alter the existing default recognition policy. The measured overhead on easy sectors is one reason to preserve that choice.

8.4 Cancellation and numerical failure

The producer's own reconstruction, source traversal, and verification loops accept cancellation checks. Calls into a third-party LP solver are checked at their boundaries; the package does not claim that a long solver call cooperatively interrupts at every internal pivot. The reproducibility notes state this limitation explicitly.

No floating-point tolerance licenses a removal. A support proposed by a numerical solution can be wrong because of scaling, degeneracy, or a tiny coordinate; the exact primal–dual conditions reject it. This makes aggressive numerical proposal generation compatible with exact downstream certificates.

9 Compressed parallelity and the missing order information

After cutting along a normal surface F, the current source constructor describes selected product handles of indices zero, one, and two. They form an interval bundle $E \to B$, together with a complementary core K. A core port is a labelled free base edge whose interval product meets a particular core chart. Original-boundary free edges are retained separately.

The base is encoded by polygon families and interval pairings. If a normal coordinate has B bits, a family can represent exponentially many polygons while storing only its binary endpoints. The existing summary records Euler characteristic, boundary count, orientability, and the set of core ports on each connected base component. It does not explicitly list the cyclic order of those ports along each boundary circle.

That order is already implicit in the complete source geometry. The problem is to extract it without expanding the polygons.

9.1 The topological information that must survive

The output needed for a replacement is more precise than a port set. It includes:

1. the cyclic sequence of port labels;
2. the entry and exit endpoint of every traversed port;
3. whether each complementary gap is a point or a nonempty arc;
4. the relative identification of the two fibre endpoints at the ports;
5. the original source chart for every contact with the core.

A zero gap means that neighboring port closures meet. Replacing it by a positive gap changes the attachment subset. A fibre reversal may not be visible in a singleton integer rank map. It must be carried as a separate bit. These two details are indispensable even when all bases are discs.

Example 9.1 (An unordered profile does not determine a marked disc)**.** Take an oriented disc with four labelled boundary ports, each separated from its neighbors by a nonempty unmarked gap. The cyclic words

$$(1, 2, 3, 4) \quad \text{and} \quad (1, 3, 2, 4)$$

have the same Euler characteristic, the same boundary count, and the same port set. No homeomorphism between the discs that preserves each port label carries one marking to the other: boundary homeomorphisms preserve cyclic order up to reversal. The example concerns the labelled relative attachment problem; it need not distinguish the unlabelled homeomorphism type of the resulting ambient 3-manifold.



Figure 1: The same four ports can have inequivalent cyclic attachment data. Green arcs are labelled ports; gray arcs are nonempty gaps. The diagrams specify cyclic order only.

10 Recovering the attaching order of compressed parallelity bundles

The existing source parallelity operator retains complete polygon charts, an inherited core with at most $30T$ handles, and at most $q \leq 66T$ labelled core ports. Its component profiles contain Euler characteristic, boundary count, fibre twisting and a set of port labels. These are not yet an explicit small attaching diagram. In particular, a set of ports does not specify their cyclic order. We give an exact extraction of this order and then a relative replacement theorem for disc products. This operation fills a specific geometric gap between a compressed cut and a small patterned handle structure; it does not assert that the resulting handle structure is admissible.

10.1 The marked boundary model

Let B be a finite, possibly disconnected, compact surface supplied as polygons with pairwise glued sides. We assume that the quotient really is a surface, including the interval-or-circle condition on every vertex link. Each unglued side lies on ∂B. Select q unglued side occurrences, with distinct labels $0, \ldots, q-1$, called *ports*. Every port carries an ordered endpoint chart $(j, 0), (j, 1)$. Endpoints of different ports may coincide after quotienting; even the two endpoints of one boundary edge cell may coincide. The statement concerns labelled edge parametrisations, and does not assume that the closure of each edge cell has distinct endpoints. All other boundary edges are called ordinary. In the source application, ordinary edges lie on the original manifold boundary.

The polygon families and their gluings are represented by affine interval pairings. Let N be the number of corner occurrences before quotienting, and let R_V be the interval equivalence relation that identifies corners across glued polygon sides. Thus the R_V-orbits are the vertices of B. Let $R_\circ$ be generated by R_V and the endpoint pairings of every ordinary boundary edge, *omitting the port edges*. In a compressed family the omission of a singleton port can be achieved by splitting that family at its rank, so q omissions create only $O(q)$ additional records. The source packet already lists each port as a singleton.

Theorem 10.1 (Punctured boundary matching)**.** *Give corner occurrence (j, e) the weight 2^{2j+e} in the first coordinate. Give exactly one endpoint occurrence of each ordinary boundary edge weight 1 in a second coordinate. Weights at coincident occurrences add. Then the weighted orbit histogram for $R_\circ$ determines all boundary circles containing ports, their cyclic port order, the entry endpoint of each port, and the number of ordinary edges in each gap between consecutive ports. No polygon or boundary-edge family needs to be expanded.*

More precisely, every orbit with nonzero first coordinate has weight

$$(2^a + 2^b,\, \ell), \qquad a \neq b, \quad \ell \geq 0,$$

and multiplicity one. These orbits give a perfect matching τ on the $2q$ endpoint tokens. Let $\iota(2j+e) = 2j+1-e$ pair the ends of each port. Alternating ι and τ gives exactly the boundary circles containing ports. The matching edge of τ carries the intervening length ℓ.

Proof. First quotient by R_V. The free edges form the ordinary boundary 1-manifold of B, a disjoint union of circles. Remove the interiors of its q port edges. Each boundary circle containing a port becomes a union of closed paths, possibly reduced to a single vertex. Every such path has exactly two incident endpoint *occurrences* from the removed edges. This remains true when consecutive ports share a vertex or a one-edge boundary circle loses its only edge. A circle without ports remains a circle and carries no endpoint mark. Interior vertices of B also carry no endpoint mark.

The connected components just described are precisely the marked $R_\circ$-orbits. Each endpoint bit was placed once, so a marked orbit has exactly the two stated bits and multiplicity one. Its second coordinate counts ordinary boundary edges because every such edge contributes exactly one unit to its component. Hence this coordinate is zero exactly for a zero-length gap; it is positive for an actual interval of ordinary boundary. Restoring each deleted port joins its two endpoints by ι, and restores the original circles. Walking these circles reads their port order and the direction of each endpoint chart. This is precisely the alternating walk in the union of the two perfect matchings. □

There are two directed alternating cycles on endpoint tokens for each unoriented boundary circle. The implementation visits each port once, rather than counting both cycles. Its canonical choice starts at the least port label on a circle, enters that port at endpoint 0, and follows the resulting direction. The endpoint labels fix this direction even when the underlying surface has no selected orientation. With $q = 0$ the output is the empty list. Unmarked boundary circles continue to be counted by the source surface-classification packet, and are not spuriously returned as marked circles.

For verification it is insufficient to compare only the sum of all endpoint integers: duplicated bits can carry in binary arithmetic. The checker requires two set bits in each nonzero mask, multiplicity one, pairwise disjoint endpoint-token sets, and their union equal to all $2q$ tokens. The underlying source-bound construction also authenticates that each mark was placed at exactly its specified source corner occurrence.

10.2 Relative fibre charts

Let $E \to B$ be the source I-bundle. Each polygon has a local product chart, and every side gluing specifies whether it reverses the fibre. These signs define the fibre-orientation double cover $\widetilde{B} \to B$. The existing source packet checks that they arise from the actual source bundle. In particular, they are compatible around vertices; arbitrary signs on a polygon adjacency graph would not suffice to define a bundle. In these source product charts the fibre transition maps are exactly $t \mapsto t$ and $t \mapsto 1-t$. Thus the recorded gauge bits determine the parametrised fibre charts themselves, not merely an abstract interval bundle isomorphism class with arbitrary fibre reparametrisations.

Theorem 10.2 (Marked fibre gauges)**.** *Place a distinct bit 2^{2j+e} at sheet e of a polygon occurrence incident to port j in $\widetilde{B}$. One weighted orbit query determines, for each port-bearing component of B, whether the fibre-orientation cover is connected. When E is a product over that component, it also determines the relative fibre reversal at every port, after the gauge at any one anchor port has been fixed.*

Proof. Over a connected component of B, the orientation cover has two connected components precisely when the bundle has a global fibre orientation. In that case each cover component

contains exactly one of the two marks of every port. Select the component containing sheet 0 of the anchor port. If it contains sheet e_j of port j, the local chart at j differs from the anchor trivialisation by the reversal bit e_j. The other component contains the complementary marks. If the cover is connected, it contains both marks of every port and no such global gauge exists. Unique port labels identify these components without any multiplicity ambiguity. □

This query is independent of the cyclic-order query. Reversing a base edge parametrisation and reversing the interval fibre are different operations. In particular, a singleton rank map does not encode fibre reversal. The implementation retains both the signed entry endpoint and the fibre bit.

10.3 A relative disc-product replacement theorem

Suppose a port-bearing component of B is a disc D. Write A for the union of its port edge cells, including their endpoints. Let $E \to D$ be the product bundle and let K be the complementary core. Its attachment to K is the specified source map from $E|_A$; endpoints shared by port cells retain their identifications and their source charts. An ordinary gap is unlabelled for this replacement, except for its status as original boundary and the retained numerical count of its source edges. The replacement does not preserve the discarded interior subdivision of such a gap as additional marked data.

Theorem 10.3 (Relative disc-product model)**.** *Let a disc component have $q_D \geq 1$ port cells. Its signed cyclic word, zero-versus-positive gap data and relative fibre gauges determine a product model with at most $9q_D$ tetrahedra, together with all port attachment charts. Replacing E by this model in $K \cup E$ gives a PL homeomorphic manifold by a homeomorphism restricting to the identity on K.*

The same homeomorphism preserves the first-cut boundary pattern when the pre-cut pattern was empty: within E this pattern is the two horizontal rims over the ordinary part of ∂D. It makes no assertion about an arbitrary unrecorded inherited pattern.

Proof. Let r be the number of positive ordinary gaps. Retain one side for every port and one side for every positive gap, inserting no side at a zero gap. Thus the new boundary cell structure has

$$m = q_D + r \leq 2q_D$$

sides. It has exactly the original cyclic order of labelled ports and exactly the original endpoint coincidences between consecutive ports. When $m < 3$, subdivide one side so that there are $n = 3$ sides; otherwise put $n = m$. Each subdivision carries its rational parameter interval in the original side chart. Hence $n = \max\{3, m\} \leq 3q_D$.

Fill the boundary polygon by a fan from one interior vertex, giving n triangles. The product of each triangle with I has a staircase triangulation into three tetrahedra. Use one global total order on the base vertices; the induced diagonals then agree on shared rectangular faces. This gives $3n \leq 9q_D$ tetrahedra.

The signed cyclic word supplies a PL homeomorphism from the model boundary circle to ∂D that agrees with every port parametrisation. On positive ordinary gaps choose any monotone PL extension between the given endpoint images. At zero gaps the common endpoint is retained. The map extends to a PL homeomorphism of discs. This is the elementary disc extension theorem; equivalently, triangulate both discs compatibly with the chosen boundary map and extend over their interiors.

Since a disc is contractible, its interval bundle is a product. Choose a global fibre trivialisation whose local gauge is the prescribed sheet 0 at the anchor port. The marked cover theorem forces

its restriction at every other port to be exactly the stored fibre bit. Taking the product with the extended disc homeomorphism therefore gives a bundle homeomorphism that agrees with every specified source rectangle chart. On a model side with parameter u and vertical parameter t, that agreement uses

$$\begin{aligned} u &\longmapsto u \text{ or } 1-u \quad \text{according to the entry endpoint,} \\ t &\longmapsto t \text{ or } 1-t \quad \text{according to the fibre gauge.} \end{aligned}$$

Restrict these maps to the stored rational subintervals when a side was subdivided. Glue this homeomorphism to the identity on K. The maps agree on every attaching chart, including coincident endpoint fibres, so the quotient map is a PL homeomorphism of the unions.

For an initially empty pattern, the new pattern inside the product is exactly $\overline{\partial D \setminus A} \times \partial I$, with the appropriate endpoint convention at the port interfaces. The constructed map preserves the ordinary boundary intervals and maps the two horizontal sheets according to the recorded gauges; hence it preserves this pattern. Any further marked data requires its own preserved charts. If r extra boundary subdivision points suffice to carry such data in product charts, the same construction uses $O(q_D + r)$ tetrahedra; one cannot retain a bound independent of r merely by omitting those marks from the input. □

Corollary 10.4 (Linear total disc-product template size)**.** *For a source triangulation with T tetrahedra, all port-bearing disc components of its source parallelity bundle can simultaneously be replaced using at most*

$$9q \leq 594T$$

tetrahedra in total, with $O(T)$ labelled port records. The core remains the same source-bound core with at most $30T$ inherited handles.

Proof. Every port belongs to exactly one component, so the sum of q_D over these disc components is at most q. The preceding relative homeomorphisms restrict to the identity on K and therefore apply simultaneously. The source bounds $q \leq 66T$ and the $30T$ core-handle bound give the estimates. □

This corollary is a bound on the added disc templates. It is not itself a tetrahedron count for the entire cut: non-disc bundle components may remain, and the core must still be triangulated compatibly with its source attaching charts. Port-free disc bundles are separate components from the core and can be retained as counted product types. Explicitly listing an enormous number of such balls would defeat compression. Most importantly, the operation replaces a disc product; it does not delete it. A ball attached along several rectangles need not be removable without changing topology.

10.4 Why the extra boundary data is necessary

Proposition 10.5 (Port profiles do not determine marked base attachments)**.** *The Euler characteristic, number of boundary components, orientability, set of port labels, and product/twisting type do not determine the labelled base of a disc product. Consequently they do not determine it in the category of bundle homeomorphisms preserving the labelled port charts.*

Proof. Take a disc with four pairwise disjoint labelled port arcs. In one model their cyclic order is $(1, 2, 3, 4)$; in another it is $(1, 3, 2, 4)$. Give the ordinary gaps equal subdivisions and choose trivial fibre charts. All listed numerical data and port sets agree. A homeomorphism of discs restricts to a homeomorphism of their boundary circles, which preserves or reverses cyclic order. The two labelled cyclic words are not related by a rotation or reversal. Hence

there is no homeomorphism preserving all port labels. A bundle homeomorphism induces a base homeomorphism and therefore cannot identify these two marked products. □

The category matters here: without the fibre structure or further boundary marking, an arbitrary homeomorphism of a 3-ball can permute disjoint disc patches on its boundary. The cyclic-order argument alone does not rule out such a homeomorphism. If the first-cut rim pattern is also retained and all port fibre gauges are fixed to zero, that pattern supplies an additional obstruction: its arcs join exactly the adjacent port endpoints. The two four-port examples have different labelled adjacency relations and cannot be identified while preserving both those rim arcs and the port charts.

Even a cyclic word needs zero-gap information. Two adjacent ports have a common endpoint fibre, whereas two separated ports do not. Their signed two-port words can agree. The total number of ordinary edges can also be made equal by subdividing the unique ordinary gap in the adjacent case. The second coordinate in Theorem 10.1 retains the gap-by-gap distinction and prevents this loss of attaching data.

10.5 Encoded complexity and an unconditional orbit-count reduction

The practical extractor uses one two-coordinate weighted query for port paths and one scalar weighted query for fibre marks, in addition to the four existing source queries. If the polygon packet has s interval records, the two new query inputs have $O(s+q)$ records. Endpoint masks have $2q$ bits; this matters for bit complexity. The marking input can use $O(q^2)$ bits even though it has only $O(q)$ records. With $L = \lceil \log_2(N+2) \rceil$ and the analogous logarithm for the polygon universe, a coarse input bound is

$$O((s+q)L + q^2).$$

The output contains $O(q)$ port and template records. Gap counts use $O(L)$ bits each. The supplied implementation's weighted replay has its existing dependence on the realized orbit trace and weight-run partitions. We do not infer a uniform polynomial bound for those partitions from a small query description.

There is nevertheless a separate deterministic polynomial construction of the *topological* port data, using only unweighted interval-orbit counting. This gives a useful distinction between the mathematical availability of a bounded extraction and the specific faster weighted implementation.

Lemma 10.6 (Orbit-membership by a singleton merge)**.** *Let $c(R)$ be the number of orbits of an interval-pairing relation R on a finite universe, and let $[x \sim y]$ denote one added singleton pairing. Then*

$$c(R \cup [x \sim y]) = \begin{cases} c(R), & x, y \text{ already belong to the same orbit,} \\ c(R) - 1, & \text{otherwise.} \end{cases}$$

Proof. The added relation identifies exactly the two existing equivalence classes containing x and y. They are either the same class or distinct classes. □

Lemma 10.7 (Orbit membership in a set by a singleton star)**.** *Let R be an interval-pairing relation and let U be a nonempty finite list of marked positions. Choose $u_0 \in U$, and let $R[U]$ be obtained from R by adding singleton pairings $u_0 \sim u$ for all $u \in U$. Then, for any position x,*

$$c(R[U] \cup [x \sim u_0]) = c(R[U]) \quad \Longleftrightarrow \quad [x]_R \cap U \neq \varnothing.$$

The construction uses at most $|U| - 1$ nontrivial added pairings, and remains valid when distinct endpoint tokens occupy the same position.

Proof. The singleton star merges exactly the original R-classes that contain a position of U, and changes no other class. Hence x is in the resulting class of u_0 exactly when its original class meets U. Apply the singleton-merge lemma to $R[U]$. Repeated positions merely give redundant or identity pairings and do not change this reasoning. □

Theorem 10.8 (Unweighted polynomial extraction)**.** *For a valid compressed polygon surface with q labelled boundary ports, their boundary matching, signed cyclic words and zero-gap flags can be computed with $O(q\log(q+1))$ ordinary interval-orbit counting calls. Product components' relative fibre gauges require at most $O(q^2)$ further calls, or $O(q)$ when the port partition into base components is supplied. Each call has $O(s+q)$ interval records and the same binary universe scale. Consequently these data have a deterministic polynomial-time extraction in s, q and the binary universe length, by the Agol–Hass–Thurston orbit counting theorem. With the supplied source component partition, all these topological chart data use $O(q\log(q+1))$ unweighted counting calls in total.*

Proof. Choose the least unmatched endpoint token x. By Theorem 10.1, exactly one other unmatched endpoint token is in its $R_\circ$-class. Bisect the remaining candidate list. For a nonempty half U, apply Lemma 10.7 to $R_\circ$, using two ordinary orbit counts. Since x is not a token of U, the answer says precisely whether its unique partner is in that half. Continue with the indicated half until one candidate remains, remove the matched pair, and repeat. Each test starts with the original relation $R_\circ$ and its own star; stars from earlier tests are not retained. There are $O(q\log(q+1))$ tests altogether and at most $O(q)$ extra singleton pairings in any one test. Coincident marked positions are handled by the same lemma. This recovers τ. For each matched pair, repeat the membership test using R_V instead of $R_\circ$. Its gap has zero ordinary edges if and only if its endpoints are the same quotient vertex. Otherwise the gap is a nonempty path. These q further tests use the simpler singleton-merge lemma and a shared baseline count for R_V. Reading the alternating matching cycles gives the signed words.

If required, partition port polygon occurrences into base components using the same membership test on the polygon-gluing relation, in $O(q^2)$ calls. For each component choose an anchor. Its two sheet marks lie in the same fibre-cover orbit exactly in the twisted case. In the product case, compare each port's sheet-0 occurrence with the anchor's sheet-0 occurrence. Equality gives gauge 0; inequality gives gauge 1, since exactly one of that port's sheets lies in the anchor component. These are $O(q)$ more tests.

All added pairings are singletons, with at most $O(q)$ of them in any instance. Apply the ordinary polynomial interval-orbit counting algorithm to each instance. The number of calls and the encoded size of each instance are polynomial in the stated parameters. Numerical gap lengths are not obtained by this argument; their zero-versus-positive flags are the information required by the relative topological replacement theorem. □

The unweighted construction is a proved reduction, not the code path benchmarked here. The implementation uses weighted extraction for fewer queries and records exact gap lengths. This distinction prevents a trace-sensitive implementation measurement from being silently upgraded into a hierarchy-wide complexity claim.

10.6 Certificates, experiments and remaining obligations

The code files are `parallelity_port_words.py` and `parallelity_port_words_verify.py`. The certificate includes the entire existing source parallelity certificate, two new independently replayable weighted-orbit proofs, signed boundary words, fibre gauges, and disc-product templates. The verifier first authenticates the source geometry using the existing independent raw-cell reconstruction. It then independently constructs the two queries, replays their proofs, decodes

the boundary and fibre information by separate code, and reconstructs the template charts and triangulations. It does not call a port-word producer helper. All six discovery queries share the cycle budget. The verification cap counts all six supplied trace lengths. An exhausted discovery returns no geometry, word, template or certificate; an exhausted replay is false.

The finite oracle in `parallelity_port_research.py` explicitly glues all polygon corners, materializes boundary edges and follows their two incident darts. A separate breadth-first propagation on the polygon adjacency graph determines fibre gauges. Thus it checks geometric order, adjacent ports and fibre data independently of weighted histograms.

The research audit also assembles the emitted tetrahedra with Regina and checks validity, connectedness, orientability, one boundary component and the exact ball-recognition predicate. This is a check of the actual emitted complexes, beyond an Euler-characteristic or homology comparison. It is not a dependency of the production extractor or certificate replay; their correctness is supplied by the explicit disc-product construction above. Every port's rational subdivision intervals are separately checked to form an exact partition of $[0, 1]$.

The tests include zero ports, one port, adjacent ports, an actual triangular disc quotient whose only boundary edge has coincident endpoints, source self-gluings, relabellings, disconnected bases, one-sided surfaces, independently replayed hostile certificates, exact binary transport of 2^{16384} source multiplicity, shared caps and cancellation. The source two-tetrahedron example with a single selected two-handle has eight adjacent ports. Its replacement has eight fan triangles and twenty-four tetrahedra; there are no original-boundary gaps to insert.

For integration into the hierarchy, the main remaining obligations are: carry an arbitrary existing pattern through these charts; supply any pattern marks lying in product horizontal faces; construct and verify the admissible patterned handle structure; control non-disc bundle components; and prove the reductions and search bounds across successive hierarchy stages. The present homeomorphism theorem preserves a supplied cut but does not decide whether a decomposition square is trivial or whether a boundary pattern is essential. It also does not assert the new small triangulation has bounded handle type suitable for every later normality condition.

11 Composition, scope, and asymptotic consequences

The two reductions act on different objects. Support reduction simplifies the search description before a normal surface is selected. Port recovery simplifies a geometric interface after the surface has been cut. Neither reduction substitutes algebraic evidence for a topological witness.

11.1 A source-preserving composition theorem

Theorem 11.1 (Compositional soundness of the two interfaces)**.** *Suppose the input triangulation and normal-coordinate conventions satisfy the validated source interfaces. Perform the following operations:*

1. *remove sector coordinates only after a complete support certificate has been independently verified;*
2. *obtain a normal surface by a certified source search in the reduced sector;*
3. *construct and verify its source cut and parallelity packet;*
4. *recover and verify port words and fibre data;*
5. *replace only certified disc-base product components, using the recorded relative charts and the pattern convention of the disc-template theorem.*

Then no normal vector in the original requested sector is lost by the first operation. Each replaced cut component is homeomorphic to its template relative to the core and the specified marked

boundary data. Every conclusion made by an unchanged valid downstream procedure on these equivalent objects remains sound.

Proof. The support theorem identifies an actual coordinate face containing every feasible vector, and restriction and zero extension give inverse linear isomorphisms of the relevant standard cones. The source search therefore returns a surface in the original triangulation. Independent cut verification fixes the geometric object to which the port data refer. The disc-template theorem supplies a PL homeomorphism on each replaced bundle whose restriction to the attachment set agrees with the retained source chart. Extend these homeomorphisms by the identity on the core and on all unreplaced pieces. They agree on intersections, so the gluing lemma gives the asserted homeomorphism of the cut object with its replacement. A valid invariant statement transports across this homeomorphism. □

The last phrase is deliberately conditional on the validity of the downstream procedure. For example, proving the cut boundary is a sphere does not prove that the component is a ball. Likewise, a standard normal coordinate vector need not satisfy every local normality condition in a patterned handle hierarchy.

11.2 The bound obtained for a supplied sector

Write L_A for the bit length of the exact source system, s for its effective quadrilateral dimension, and V for the work of essential-disc tests and their certificates. Let $\mathsf{LP}(L_A)$ denote the actual support-producer cost. The support-and-enumerate route has the form

$$\mathsf{LP}(L_A) + \operatorname{poly}(L_A) + (1+k)^{O(s)} \operatorname{poly}(L_A) + V.$$

The first term is present even if support is unchanged. In the mathematical algorithm using a polynomial-time rational LP routine, it is polynomial in L_A. The production solver in this package is not assigned that worst-case bound.

If $s = O(\log N)$, $k, L_A \le N^{O(1)}$, and the remaining certified tests have comparable cost, this *supplied-sector task* is quasi-polynomial in N. If the geometry falls within the existing specialized dimensions $s \le 3$, its enumeration is substantially more concrete. Neither statement controls how many sectors must be considered to find a disc.

11.3 The bound obtained for geometric replacement

Let P be the encoded polygon-pairing size, W its represented integer universe, and q the number of singleton core ports. The endpoint-orbit formulation yields polynomial reductions in

$$P + q + \log(W+1)$$

to ordinary interval-orbit counting. The fast implementation uses weighted queries and is charged for its realized exact traces and weighted partitions. Its emitted disc templates have total combinatorial size $O(q)$ for port-bearing disc components, in addition to the inherited core and source provenance.

At the inspected checkpoint, $q \le 66t$ and the inherited core has at most $30t$ handles. These are bounds from the preceding source parallelity construction, not newly proved global hierarchy bounds. The present theorem explains how to preserve the marked contacts when using those bounds.

A component with no core port requires separate treatment. A disc product is then an isolated ball component relative to this decomposition. One may retain a single template and binary

multiplicity when the specified marks are identical; arbitrary source-boundary marks or component identities cannot be discarded. Non-disc bases retain their original compressed bundle representation.

11.4 What is still required for the desired recognizer

A sufficient global theorem would need all of the following, with the parameters referring to an input diagram of n crossings:

1. a source binding from the diagram to the exterior and its peripheral data, with polynomial encoded size;
2. a complete family of candidate sectors, charts, or hierarchy surfaces whose total search work is quasi-polynomial;
3. controlled effective dimensions or another finite search parameter;
4. admissible patterned handle structures after each transformation, including every inherited pattern arc and each required normality condition;
5. a bound on the total number of successful and failed transitions, restarts, and certificate operations;
6. a bound on encoded state growth, together with valid positive and negative terminal tests.

The new interfaces help with items 3, 4, and 6. They do not establish items 2 or 5. In particular, the one-column revival example prevents a naive claim that nearby sectors automatically inherit a small effective dimension. The disc-template theorem also cannot be applied to an annulus or a nonorientable base just because their ports are easy to list.

11.5 A precise conditional global search budget

Theorem 11.2 (Search-tree budget)**.** *Let an algorithm on an input of size n have a finite rooted search tree $\mathcal{T}$, including rejected children, repeated states, and restarts. Write $b(v)$ for the number of children of v, and let $H(n)$ bound the number of edges on every root-to-leaf path. Suppose that every such path P satisfies*

$$\sum_{v\in P} \log \max\{1, b(v)\} \le B(n).$$

If all work is chargeable to tree nodes or incoming edges at cost at most $W(n)$ per node, then

$$\#\,\mathrm{leaves}(\mathcal{T}) \le e^{B(n)}, \qquad |\mathcal{T}| \le 1 + H(n)e^{B(n)},$$

and the total cost is at most $W(n)\big(1 + H(n)e^{B(n)}\big)$. In particular, polynomial H, W and $B(n) = O((\log n)^2)$ give $n^{O(\log n)}$ time; $B(n) = O((\log n)^3)$ gives $n^{O((\log n)^2)}$ time.

Proof. Give the root mass 1, and divide the mass of every internal node equally among its children. Since the tree is finite, the leaf masses sum to 1. The mass of a leaf is the reciprocal of the product of the positive branching numbers along its path, and is therefore at least $e^{-B(n)}$. This proves the leaf bound. Every nonroot node lies on a root-to-leaf path, so the union of these paths contains at most $H(n)$ nonroot nodes per leaf. Summing the assumed node charges proves the cost bound. □

Here $W(n)$ includes state construction, exact arithmetic, surface tests, certificate production and replay, and the generation of children. Generating a large child list may instead be charged incrementally to the respective child nodes. Failed surface candidates and unsuccessful LP attempts must either occur explicitly in the tree or be covered by these local charges. A sequence of unary restarts has zero branching charge but consumes depth; the hypothesis on H prevents it from being free. In particular, polynomial encoded state size alone does not bound the work of the present simplex backends.

Corollary 11.3 (Charging dimension decreases and their replenishment)**.** *Suppose a nonnegative integer $D(v)$ is assigned to every tree node. Designate certain binary branching nodes as conflict nodes, and assume D decreases by at least one on either outgoing edge of every such node. For a path P, let*

$$R(P) = \sum_{\substack{(v,w)\in P \\ v \text{ not a conflict node}}} \max\{0, D(w) - D(v)\}, \qquad C(P) = \sum_{\substack{v\in P \\ v \text{ not a conflict node}}} \log \max\{1, b(v)\}.$$

Then its branching budget is at most

$$(\log 2)(D(\text{root}) + R(P)) + C(P).$$

Proof. If the path traverses r conflict branches, telescoping the changes in D gives $0 \le D(\text{leaf}) \le D(\text{root}) - r + R(P)$. Each conflict node contributes $\log 2$; the other contributions sum to $C(P)$. □

The intrinsic-dimension conflict theorem supplies the required decrease within a fixed matching problem. Replacing a sector, changing the source, or restarting a hierarchy can replenish the dimension; the positive increases must be charged through $R(P)$, or through a separate proved phase budget. This is a sufficient accounting target, not a bound on $R(P)$, $C(P)$, or hierarchy depth. To obtain a recognizer one must also prove that the search rules cover every required possibility and that all terminal conclusions are valid; neither the counting lemma nor a capped or inconclusive leaf establishes those topological facts.

12 Implementation evidence and measured performance

The evidence is divided into exact mathematical checks, finite geometric oracles, and paired timing experiments. These answer different questions. An exact certificate establishes a statement about its supplied input; a finite audit checks an implementation on the audited cases; elapsed times describe this machine and protocol. None substitutes for a worst-case theorem.

12.1 Environment and timing conventions

The recorded environment is Python 3.12.14 on Linux, with SciPy 1.17.0, SymPy 1.14.0 and Regina 7.4 (distribution version 7.4.1). The experiments use the pinned repository checkpoint specified in Section 1. Exact source pairings, random seeds, selected supports, implementation hashes, all timing samples and retained certificates are included in the package. The two final timing campaigns ran separately, with the other research workers idle.

The support campaign uses nine shuffled paired repetitions after one warmup per arm; the geometric campaign uses seven. Python's ordinary garbage collection and allocation remain enabled. Imports are warm. In the tables, individual time columns are medians, while a column labelled "paired ratio" is the median of the ratios within the paired trials. It need not equal the ratio of the displayed medians. This distinction is material in several noisy small cases. All samples, rather than only the summary statistics, are retained.

12.2 The maintained source-sector audit

The support audit starts from the maintained `reports/57/results/discovery_corpus.json`. It constructs sectors from archived standard and quadrilateral supports, small allowed sets, and reproducibly sampled full supports. Among the resulting sectors, 188 have raw matching

dimension greater than three. These are the cases tested by the higher-dimensional audit; they are not all sectors of all triangulations in the repository.

For every one of these 188 cases, the exact support certificate agrees with the support union of all archived compatible standard vertices. The raw dimensions are four or five. In 24 cases, removing forced zeros lowers the dimension; the resulting effective dimensions are three, four or five. However, the incumbent two-row matching closure already finds the same forced coordinates in every audited case. Thus this corpus validates the new complete certificate, but supplies no example of a dimension advantage over that inexpensive incumbent reduction.

The initial three-repeat complete-query pilot was slower with the new route on all five selected small sectors. It is retained as pilot evidence. The more carefully paired final measurements below supersede its timing estimates.

12.3 What the abstract obstruction tests establish

The cyclic family from the theorem was tested at $m = 3, 8, 16, 32, 64$, with effective dimensions one and three. All exact support proofs replay successfully. In every case, the incumbent one-/two-row closure makes no reduction, while the complete certificate removes every p-coordinate. The reduction in raw versus effective dimension is a proved property of the whole infinite family, not an extrapolation from these ten tests.

m	Raw d	Effective s	Pair zeros	Exact zeros	Raw basis candidates
3	3	1	0	3	15
8	8	1	0	8	11,440
16	16	1	0	16	$\approx 5.657 \times 10^{8}$
32	32	1	0	32	$\approx 1.777 \times 10^{18}$
64	64	1	0	64	$\approx 2.358 \times 10^{37}$

Table 2: The abstract cyclic family with effective dimension one. Large candidate counts are rounded here and exact in the CSV/JSON.

The last column is the combinatorial number $\binom{2m}{m-1}$ of candidate coordinate-hyperplane bases in the raw dimension for the displayed $s = 1$ family. It is an operation-space comparison, not a measured elapsed time or a lower bound for every possible algorithm. The benchmark does not attempt to enumerate those enormous raw arrangements. After support restriction, the cone is one ray. The JSON also records the $s = 3$ cases and the exact certificates.

This separation is relevant to asymptotic design, but the displayed matrices have not been realized as an infinite family of normal matching systems of knot exteriors. The separate geometric search addresses that gap experimentally without claiming to close it theoretically.

12.4 A genuine geometric example, and an important negative search

A deterministic bounded search examines larger layered, capped and Pachner-modified solid tori. The first run, seeded by `maximal-support-geometric-20261012`, stops after finding a counterexample to one fixed-row closure pass. It tests 2,187 sampled sectors in 18 sources; their raw dimensions range from one to ten. Among the 794 cases of raw dimension at least four, 683 complete support checks need no LP proposal and 111 use one.

The discovered source has 36 tetrahedra and 36 allowed quadrilateral types. Its raw dimension is eight. One incumbent matching-support pass forces 22 coordinates to zero; the complete certificate forces 23, including allowed-coordinate index zero. The full source pairings, move history, support and certificate are supplied as a regression fixture.

There is a crucial limitation: after the first 22 coordinates have been removed and the kernel rebuilt, its raw dimension is already five, equal to the complete certificate's effective dimension. A second incumbent pass on that rebuilt kernel finds the additional zero. Thus this is a genuine fixed-pass incompleteness example, but not an effective-dimension separation from iterated elimination and two-row closure.

The second search explicitly repeats those inexpensive reductions after each rebuild. It completes the scheduled 4,864 sectors in 38 sources, including 1,221 sectors of dimension at least four. Of those, 993 obtain an exact complete-support confirmation without a new LP proposal and 228 use an LP proposal. It finds no additional forced zero beyond the iterated incumbent procedure. This is useful negative evidence for prioritizing inexpensive iteration. It is not a theorem that iterated two-row closure is complete on geometric matching systems. In particular, no finite search can establish the missing infinite-family realization or exclusion theorem.

12.5 Final supplied-sector timing results

Both timing arms perform a complete standard supplied-sector call and return the same status on every case. The new arm includes support production, exact reconstruction, its independent source replay, rebuilding when needed, and ordinary downstream discovery. The incumbent uses its existing discovery route. All checking performed internally by the queried APIs is included. A separate final replay by the unchanged witness or exhaustion checker is performed outside the measured calls. This external replay is not included in the timings for either arm.

Two scopes are reported. "Fresh source" includes source preparation within each call. "Prepared source" gives both arms the same reusable `PreparedSectorSource`; it therefore tests the combined reduction and discovery routes after common source compilation. The latter comparison helps separate preparation cost from later work, but is still not an isolated LP microbenchmark.

Case	t	Incumbent (ms)	New (ms)	Paired new/old
`cap_3_5_2`	5	6.52	8.41	1.355
`cap_3_5_3`	6	32.99	37.27	1.088
`cap_1_2_3`	4	13.15	19.37	1.036
LST9, mixed	9	7.41	8.27	1.103
LST9, meridian	9	2.96	4.01	1.328
Pachner36	36	890.44	537.90	0.604

Table 3: Complete supplied-sector calls with fresh source. Nine paired trials; lower paired new/old is favorable. Time columns are individual medians. LST9 denotes the nine-tetrahedron Fibonacci layered source.

Case	t	Incumbent (ms)	New (ms)	Paired new/old
`cap_3_5_2`	5	5.41	7.25	1.418
`cap_3_5_3`	6	30.55	33.55	1.091
`cap_1_2_3`	4	15.10	15.56	1.030
LST9, mixed	9	8.74	9.42	1.078
LST9, meridian	9	2.54	3.81	1.408
Pachner36	36	875.82	513.11	0.573

Table 4: Complete supplied-sector calls with shared prepared source. Nine paired trials; lower paired new/old is favorable. Time columns are individual medians. LST9 denotes the nine-tetrahedron Fibonacci layered source.

The 36-tetrahedron fixture improves in every one of its nine paired trials in both scopes. Fresh-source median time falls from 0.890441 seconds to 0.537895 seconds, with paired new/old ratios

ranging from 0.535 to 0.662. With a shared prepared source, the medians are 0.875823 and 0.513108 seconds, and the paired ratios range from 0.520 to 0.698. The identical-incumbent control for this fixture has median paired ratio 1.002. These results support a substantial complete-query gain on this particular geometric fixture.

The gain does not arise from a smaller final arrangement: both routes begin 1,835 bases and emit 31 rays across their quadrilateral and standard phases. Their final effective dimension is five. The result demonstrates reduced implementation cost for the combined support route, not an improved exponent or a proven dimension advantage.

The five smaller cases remain unfavorable to the new route under the paired median statistic. Some effects close to one are comparable to measurement variability: the identical-code controls on these small cases have median paired ratios between 0.922 and 0.981. The complete sample arrays also show why one should not attach undue meaning to a few percent on a millisecond-scale call. The evidence supports retaining an explicit or adaptive invocation policy rather than adding a mandatory LP to every sector query.

12.6 Finite and binary geometric audits

The final port audit comprises 129 finite cases and eight binary cases. It includes the retained source proof bank, analytic layered sources, source relabellings, zero surfaces, and large multiplicities. The finite oracle explicitly glues polygon corners, constructs boundary darts, walks them, and propagates fibre signs on a separate adjacency graph. It therefore does not derive its answer from the new weighted histograms. All finite words and product fibre gauges agree.

Audited quantity	Count
Finite source cases	129
Binary source cases	8
Explicitly traversed boundary edges	10,178
Port occurrences	4,048
Zero gaps / positive gaps	3,768 / 280
Reversed port traversals	1,408
Nonzero product fibre gauges	897
Twisted port-bearing components	20
Emitted attached disc templates	329
Tetrahedra across these templates	10,218
Individually checked port subdivision charts	3,067

Table 5: Counts from the completed geometric audit. These are totals over cases, not a single simultaneous cut.

There are 21 distinct emitted product complexes, with between nine and 330 tetrahedra. Every one passes Regina's validity, connectedness, orientability, one-boundary-component and `isBall()` checks. Thus the finite validation uses actual ball recognition, in addition to the product-construction proof; a homology calculation or sphere boundary test alone would not justify that conclusion. The audit also verifies that the rational port subintervals cover $[0, 1]$ exactly, with the intended orientations and no overlapping interiors.

Binary tests include source multiplicity 2^{16384}, both with no ports and in a two-tetrahedron source with two attached ports. These are exact encoded-input tests, not literal expansion experiments. The latter attached example returns two nine-tetrahedron disc templates. The archive retains 86 complete source certificates, selected across distinct audited profiles and all binary cases.

Hostile-certificate tests include altered words, incorrect gauge bits, changed source data, malformed masks, and invalid template charts. The independent verifier is also tested with producer and query-building helpers disabled. Shared cycle caps, zero budgets and cancellation produce the specified incomplete outcomes rather than accepted partial geometry.

12.7 Paired compressed-versus-literal geometric timings

The comparison asks both arms to recover the same signed port words, relative fibre gauges and disc-product templates from the same supplied normal surface. Both construct and independently replay the source parallelity packet. The compressed arm additionally produces and replays the two new weighted-orbit proofs. The literal arm expands the boundary, walks its darts and propagates the fibre orientations, then constructs the same templates. Both serialize their common returned word, gauge and template data inside the measured call; the serialized outputs are byte-identical and their hashes are retained. This does not claim that the two different proof encodings are identical.

t	Multiplier	Compressed (ms)	Literal (ms)	Paired literal/compressed
2	4	45.01	28.98	0.639
2	64	49.51	36.37	0.729
2	1,024	50.83	188.59	2.749
2	4,096	65.84	885.90	13.455
4	64	333.50	253.67	0.759

Table 6: Matched supplied-surface port recovery, including source construction, proof replay and common-output serialization. Seven paired trials. A paired ratio above one favors compression.

On the two-tetrahedron family, increasing multiplicity from four to 4,096 increases the represented boundary while the common output stays at 1,250 serialized bytes. At multiplicity 4,096 the compressed median is 65.84 ms, compared with 885.90 ms for the literal arm; the median paired literal/compressed ratio is 13.455. All seven pairs favor compression. Its time still increases with input bit length and realized trace work; the experiment does not assert constant running time.

At small multiplicities the additional queries and proof replay cost more than the saved traversal. The four-tetrahedron case at multiplicity 64 also favors the literal arm. The identical-compressed-code control has median paired ratio 1.019. The multiplicity-1,024 measurements show substantial allocation/host variability, which is visible in the raw samples and in the difference between paired ratios and ratios of medians. We report the observed crossover and large-case improvement, without treating this five-row table as a worst-case scaling proof.

12.8 Regression status and practical conclusions

The final combined regression command runs 59 tests successfully. It includes the 14 new support tests and ten new port tests, together with the relevant existing matching-support, sector-support, sparse kernel, normal-sector integration and source-parallelity suites. The exact command and output are saved in `validation/regression-tests.txt`. This is a focused regression run, not a claim to have executed every test in the repository.

The implementation evidence suggests two immediate policies. First, retain cheap forced-zero closure and consider rebuilding and repeating it before paying for a full support LP; invoke the complete producer when dimension, repeated heuristic work or certificate reuse makes it plausibly worthwhile. The precise crossover needs broader complete-task measurement. Second, use the compressed port route for large represented geometry while retaining the possibility of

direct traversal on genuinely small objects. Both decisions should be made from encoded size and observed work, with exact verification after either route.

These policies can improve current performance while the structural questions remain open. The abstract separation explains a possible asymptotic benefit; the source experiments establish actual correctness and selected runtime gains; the global search-budget theorem specifies what still has to be bounded before either becomes a quasi-polynomial unknot-recognition theorem.

13 Integration decisions and corrections to tempting arguments

13.1 The support rule should initially remain explicit

Exact complete support can be valuable when raw dimension overstates feasible dimension. It can be pure overhead when the cheap matching preprocessor already finds every forced zero, or when the original sector is already easy. The opt-in wrapper makes this cost visible. A later policy can use raw dimension, the result of cheap propagation, and historical support-certificate reuse to decide when to invoke the new producer. Such a policy must be tested on complete calls, including verification and recompilation.

A support certificate is reusable mathematical information. It should be cached with its source digest and allowed column set, not just with the retained dimension. An incompatible sector, changed face gluing, or changed column ordering invalidates that binding.

There is a further implementation detail behind the reuse theorems: their column sets belong to a fixed matrix. A sector-dependent elimination may change the displayed row basis. A cached multiplier must therefore be transported back to the original source equations, or represented in one fixed source constraint system, before evaluating a newly allowed column. Padding an unrelated reduced row vector with zeros is not a valid transport operation.

13.2 Templates are a relative replacement, not deletion

A disc bundle attached to the core can change the topology of the union. Its being homeomorphic to a ball does not authorize removing it. The emitted template is glued using the same labelled relative attachment data. This is why cyclic order, gap incidence, and fibre gauges are part of the result.

The initial pattern convention is equally important. The theorem preserves the horizontal/vertical seam marking arising from a first cut with no additional old pattern inside the bundle. For a later hierarchy step, inherited marked curves can wind through the base. Those curves must be transported explicitly or supplied in an additional compressed pattern representation. The package does not silently erase them.

13.3 Effective dimension is not a numerical rank estimate

All reported exact dimensions arise from rational matrix ranks and verified support. A floating-point rank or a tolerance-based count of small coordinates is inadequate in a certificate path. This is especially relevant to layered families, where integral normal coordinates grow quickly even though their encoded descriptions are short.

13.4 How to interpret the obstruction examples

The cyclic sparse matrix family is an unconditional obstruction to bounded-row sign inference for general rational matching systems. It uses small coefficients and sparse rows. We do not assert

that every member is the quadrilateral matching matrix of a triangulated 3-manifold. Geometric realization would be an additional theorem.

Similarly, the one-column revival example establishes that an algebraic dimension-continuity argument is false without its dual sign hypothesis. It does not establish that every geometric sector neighborhood has such a jump. This distinction prevents an abstract hardness or counterexample from being promoted into a topological impossibility claim; the same care is required when applying the abstract hardness results of Burton and He [8].

14 Further research questions and concrete next experiments

The most useful next work is to connect the proved interfaces to structural theorems about triangulations. The following questions are ordered roughly by how directly they continue the present implementation.

Research problem 14.1 (A solver with a controlled exact running time)**.** Can the maximal-support producer use a practical rational LP method with a suitable polynomial bit-complexity guarantee, while retaining the current small independent checker? Compare exact simplex, rational reconstruction of interior-point solutions, and support-only separation methods on the same source corpus. Compare the sparse original-source formulation of Corollary 8.1 with elimination into quadrilateral coordinates, measuring coefficient growth and total certificate-production cost. The question concerns the producer; the certificate format already supplies the proof interface.

Research problem 14.2 (Certifiable support reuse over sector neighborhoods)**.** Implement the quotient-extension and dual-activation theorems as persistent search data. Measure how often dual sign tests dispose of an extension without a new LP, and how many quotient coordinates remain. For replacements, identify precisely which part of the old positive support survives; an extension theorem cannot be reused after deleting its positive anchor.

Research problem 14.3 (Geometric realization of hidden forced zeros)**.** Do triangulated knot exteriors realize families in which every useful nonnegative row consequence requires an unbounded number of source rows, while effective dimension stays bounded? Either a construction or a geometric obstruction would explain when complete support reduction can change asymptotic behavior rather than just constant factors.

Research problem 14.4 (Small effective dimension from topology)**.** Which explicit classes of triangulations admit a complete family of useful sectors with effective dimension $O(\log n)$, even when every meridian uses $\Omega(t)$ tetrahedra? Dense quadrilateral support and small feasible dimension are compatible. The objective is a topological coverage theorem, not an assumption about the successful sector.

Research problem 14.5 (Objective-aware conflict branching)**.** Implement the dimension-controlled feasibility search with certificate reuse. Determine which triangle gauges, homological constraints, or boundary conditions give a useful linear target while preserving essential-disc coverage. Positive Euler characteristic alone must not be substituted for essentiality.

Research problem 14.6 (Patterned disc-bundle replacements)**.** Extend the port-word interface to a compressed marked graph embedded in the disc base or on its boundary. Preserve trivalent vertices, cyclic incidence, and source charts. Establish a replacement bound in the number of genuinely distinct marks rather than the number of represented normal discs. Existing fast curve algorithms [9] are a natural external primitive.

Research problem 14.7 (Annular and higher-genus base templates)**.** For annular bases carrying inherited marked arcs, develop a compressed relative-coordinate system that transports the

entire marked pattern through a replacement. A chosen transverse reference arc permits winding integers; record their changes under Dehn twists and distinguish such reference-dependent coordinates from invariants under the permitted boundary-preserving homeomorphisms. Without a fixed interior reference or required isotopy class, a Dehn twist is itself a boundary-fixed homeomorphism and is not an obstruction to unpatterned annulus classification. Determine whether the needed integers and subdivision data remain polynomially encoded through the hierarchy. For higher genus, construct a compressed cut graph with explicit source charts and prove that it transports every inherited mark.

Research problem 14.8 (Admissibility after relative replacement)**.** Construct the characteristic surface and boundary pattern of the resulting handle structure, then verify the actual admissibility conditions. The existing one-tetrahedron empty-cut obstruction shows that bounded inherited handle count does not imply admissibility. A reduction must either return the required admissible structure or a correctly transported violating disc.

Research problem 14.9 (A deterministic cost bound for the used weighted engine)**.** The unweighted orbit reduction gives a clean polynomial theoretical route. The current faster implementation accounts for realized weighted partitions and traces. Prove a uniform bound for the particular port-endpoint weight systems, whose nonzero endpoint masks occur only q times, or exploit their special two-endpoint structure to avoid general weighted refinement.

Research problem 14.10 (Reusing unaffected hierarchy components)**.** After a compression or admissibility repair, which later pieces retain valid source and attachment certificates? Build a dependency relation with a geometric persistence proof. Then bound total recomputation using that relation. Merely caching a syntactically similar object does not prove that its source manifold and pattern are unchanged.

Research problem 14.11 (A genuinely global complexity potential)**.** Find a potential that charges sector discovery, unsuccessful LP calls, cutting, pattern transport, and hierarchy restarts together. A useful target is a total logarithmic state budget of $O((\log n)^2)$ for an $n^{O(\log n)}$ bound, or $O((\log n)^3)$ for the broader announced quasi-polynomial scale. The potential must charge failed searches too.

Research problem 14.12 (Broader complete-task measurement)**.** Move from controlled source families to a larger collection of hard unknot diagrams. Record diagram conversion, positive and negative searches, certificate verification, and fallback work in one budget. Retain regressions and inconclusive cases. Use paired timing with identical-code controls before adopting an automatic policy.

15 Assessment

The concrete algebraic gain is the ability to certify the feasible coordinate face before enumerating its rays. The new certificate makes the effective dimension usable earlier and provides reusable dual information for neighboring searches. The concrete geometric gain is a small, explicit relative description of disc products, including the attachment details that an unordered topological profile omits.

Both developments replace an implicit or expensive intermediate step by an exact checkable interface. Their proofs identify useful parameters and also show why several appealing shortcuts fail. The remaining challenge is to prove that these parameters stay controlled in a complete hierarchy for every unknot exterior.

A Reproduction and artifact map

The ZIP contains the article sources and PDF, the new Python modules and tests arranged as an overlay for the inspected repository, a patch, raw JSON evidence, environment details, and reproduction instructions. The source hashes bind the experiments to the relevant baseline and new modules. The article does not require network access to compile: run `latexmk -pdf article.tex` in the article directory.

The code overlay is designed for the inspected `fast/` package. Apply the patch only after checking it against the target checkout. The exact commands and dependencies in `README.md` supersede abbreviated examples in the prose. Raw evidence distinguishes source audits, abstract matrix cases, operation counts, kernel timings, and complete wrapper timings.

A later compiler or Python version may change execution times. Mathematical assertions in the checkers are exact. Floating-point solutions are only proposals, and a failed or interrupted proposal cannot create a mathematical exclusion.

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